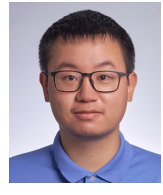
A Differential Equation Approach to the Most-Informative Boolean Function Conjecture



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26 June 2025

Mutual Information Conjecture (Kumar-Courtade '12)

Conjecture (Mutual Information)

Let $\mathbf{X} \sim \mathrm{Unif}(\mathbb{H}^n)$ *be a random variable distributed uniformly on the Boolean Hypercube and* $\mathbf{Y}$ *be received after passing each bit of* $\mathbf{X}$ *through a BSC channel with cross-over probability* $p = \frac{1-\rho}{2} \in [0,1]$*. Let* $f : \mathbb{H}^n \mapsto \mathbb{H}$ *be a Boolean function that maps a Boolean sequence to a binary value. Then the following inequality holds:*

$$I(f(\mathbf{X}); \mathbf{Y}) \leq 1 - H_2(p),$$

where $H_2(p) = -p\log_2(p) - (1-p)\log_2(1-p)$ *is the binary entropy function.*

Background

The Mutual Information Conjecture is widely studied in the Information Theory community.

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Mutual Information Conjecture (Kumar-Courtade '12)

The conjecture also has a strengthened version using Squared-Hellinger distance.

Conjecture (Hellinger)

Under the same assumptions as in the Mutual Information Conjecture, the following holds:

$$\sqrt{1-\mathbb{E}\left[f(\mathbf{X})\right]^2}-\mathbb{E}\left[\sqrt{1-\left((T_\rho f)(\mathbf{Y})\right)^2}\right]\leq 1-\sqrt{1-\rho^2},$$

where $T_\rho f(\boldsymbol{y})=\mathbb{E}\left[f(\mathbf{X})|\mathbf{Y}=\boldsymbol{y}\right]$.

Remarks

The conjectures in their limiting case $\rho \to 1$ are related to isoperimetric inequalities

- Consider $f = 1_A$, where $A \subset \{-1, +1\}^n$
- $\mathrm{Sen}_f(x)$ is the number of neighbours of x that "cross the set boundary".
  - if $x \in A$, then number of neighbors $\notin A$
  - if $x \notin A$, then number of neighbors $\in A$
- f is said to be balanced if $|A| = 2^{n-1}$.

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- Mutual information conjecture: Harper's isoperimetric inequality $\mathbb{E}\left[\mathrm{Sen}_f(X)\right] \geq 1$, for balanced f

- Hellinger conjecture: Talagrand's conjectured isoperimetric inequality $\mathbb{E}\left[\sqrt{\mathrm{Sen}_f(X)}\right] \geq 1$, for balanced f

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  - Best result along these lines $\mathbb{E}\left[\mathrm{Sen}_f^{\alpha}(X)\right] \geq 1$, for balanced f. Whenever $\alpha \geq 0.50057$ (Durcik, Ivanisvili, and Roos '24)

Background

Conjecture MI has been studied widely in the area of Noise Stability and Non-Interactive Correlation Distillation (NICD) problems.

The Mutual Information Conjecture has been established for various parameters

- Latest: $\rho \leq 0.914$, Balanced case (Lei Yu '24)

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Motivation: There is a natural semigroup structure for the noise operator Therefore, "intuitively", proving it at $\rho \approx 1$ should be the hardest. (Results are consistent with the intuition).

However as $\rho \to 1$, this is the known Harper's isoperimetric inequality.

Therefore, something seems amiss. This motivated us to look at the problem differently.

Key takeaway from this talk

Modulo four explicit inequalities that numerical simulations (and more) indicate are true

- Three of them are of the form $f_i(u, v) \geq 0$, for explicitly defined functions and $(u, v) \in [0, 1]^2$ (can easily plot the surface numerically)
- One of them involves four variables. This has been tested by grid search exhaustively

We can establish the mutual information conjecture for balanced functions (and more).

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We can establish the mutual information conjecture for balanced functions (and more).

The approach is different from the existing ones and involves differentiation along the semigroup.

Differential Equation Approach

Definition

Consider the same setting as previous, and let

$$p_t = \frac{1 - e^{-2t}}{2}$$

be the crossover probability for some $t \in [0, \infty)$, i.e., $\rho_t = e^{-2t}$. Let

$$\mathbf{Y}_t = (Y_{t,1}, Y_{t,2}, \cdots, Y_{t,n})$$

be the output of the BSC channel with crossover probability p_t. Take some arbitrary $P_{F|\mathbf{X}}$ where $F \in \{-1, 1\}$ is a binary random variable such that $P_{F|\mathbf{X}}(F = -1|\mathbf{X} = \boldsymbol{x}) \in (0, 1)$ for all $\boldsymbol{x}$. Define

$$\gamma(t) = H(F|\mathbf{Y}_t).$$

Differential Equation Approach

Remark

- *The filtration gives us the degraded structure, i.e.* $F \to \mathbf{X} \to \mathbf{Y}_{t_1} \to \mathbf{Y}_{t_2}$ *holds for any* $t_1 < t_2$.
- *By definition* $\gamma(0) = H(F|\mathbf{X})$, $\gamma(\infty) = H(F)$, $I(F;\mathbf{X}) = \gamma(\infty) - \gamma(0)$ *and* $I(F;\mathbf{Y}_t) = \gamma(\infty) - \gamma(t)$, *we have that* $\gamma(t) = H(F) - I(F;\mathbf{Y}_t)$.

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Lemma

$$\frac{d\gamma(t)}{dt} = \frac{1}{2^n} \sum_{\boldsymbol{x} \sim \boldsymbol{y}} (v_{\boldsymbol{x}}(t) - v_{\boldsymbol{y}}(t))(J(v_{\boldsymbol{y}}(t)) - J(v_{\boldsymbol{x}}(t))),$$

where $v_{\boldsymbol{x}}(t) = Pr(F = -1|\mathbf{Y}_t = \boldsymbol{x})$ *for* $\boldsymbol{x} \in \mathbb{H}^n$. *Here* $\boldsymbol{x} \sim \boldsymbol{y}$ *stands for the Hamming distance* $d_H(\boldsymbol{x}, \boldsymbol{y}) = 1$ *and the tuple* $(\boldsymbol{x}, \boldsymbol{y}), (\boldsymbol{y}, \boldsymbol{x})$ *are only counted once in the summation.*

Differential Equation Approach

Definition

Let Ψ be the class of all non-negative functions $\psi(a,b) : (0,1)^2 \mapsto \mathbb{R}$ that satisfy the following two conditions:

- $\psi(a,b) = 0$ when $H_2(a) \leq b$.
- $\psi(1-a,b) = \psi(a,b)$.
- Let P_X be an arbitrary distribution on $\mathcal{X} = \{1,2,3,4,5\}$, and $(u_x, w_x) \in (0,1)^2$ for $x \in \mathcal{X}$ be arbitrary. Then, the following holds:

$$\frac{\mathbb{E}_X[(u_X - w_X)(J(w_X) - J(u_X))]}{2} \geq \psi\left(\frac{\mathbb{E}[u_X + w_X]}{2}, \frac{\mathbb{E}[H_2(u_X) + H_2(w_X)]}{2}\right) - \frac{\psi(\mathbb{E}[u_X], \mathbb{E}[H_2(u_X)]) + \psi(\mathbb{E}[w_X], \mathbb{E}[H_2(w_X)])}{2}.$$

The cardinality bound follows from imposing Caratheodory theorem on $P_{\mathbf{X}}$.

Differential Equation Approach

Theorem

For every $\psi \in \Psi$ *we have*

$$\frac{d\gamma(t)}{dt} \geq \psi(\Pr[F = -1], \gamma(t)).$$

Consequently, if we let

$$g(x) = \int_{\gamma(0)}^{x} \frac{du}{\psi(\Pr[F = -1], u)}, \qquad \forall x \geq \gamma(0).$$

we obtain

$$\gamma(t) \geq g^{-1}(t).$$

Differential Equation Approach

Proof idea similar to Talagrand.

The proof is done by induction on n.
By Lemma (1), it is equivalent for us to show

$$\frac{d\gamma(t)}{dt} = \frac{1}{2^n} \sum_{\boldsymbol{x} \sim \boldsymbol{y}} (J(v_{\boldsymbol{y}}(t)) - J(v_{\boldsymbol{x}}(t)))(v_{\boldsymbol{x}}(t) - v_{\boldsymbol{y}}(t)) \geq \psi(\Pr(F = -1), \gamma(t)),$$

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Observe the following fact

$$\begin{aligned}\frac{d\gamma(t)}{dt} &= \frac{1}{2^n}\sum_{\tilde{\boldsymbol{x}}}(J(v^+_{\tilde{\boldsymbol{x}}}(t)) - J(v^-_{\tilde{\boldsymbol{x}}}(t)))(v^-_{\tilde{\boldsymbol{x}}}(t) - v^+_{\tilde{\boldsymbol{x}}}(t)) \\ &+ \frac{1}{2^n}\sum_{\tilde{\boldsymbol{x}}\sim\tilde{\boldsymbol{y}}}(J(v^-_{\tilde{\boldsymbol{y}}}(t)) - J(v^-_{\tilde{\boldsymbol{x}}}(t)))(v^-_{\tilde{\boldsymbol{x}}}(t) - v^-_{\tilde{\boldsymbol{y}}}(t)) + \frac{1}{2^n}\sum_{\tilde{\boldsymbol{x}}\sim\tilde{\boldsymbol{y}}}(J(v^+_{\tilde{\boldsymbol{y}}}(t)) - J(v^+_{\tilde{\boldsymbol{x}}}(t)))(v^+_{\tilde{\boldsymbol{x}}}(t) - v^+_{\tilde{\boldsymbol{y}}}(t)).\end{aligned}$$

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Using induction, we obtain

$$\begin{aligned}\frac{d\gamma(t)}{dt} &\geq \frac{1}{2^n}\sum_{\tilde{\boldsymbol{x}}}(J(v^+_{\tilde{\boldsymbol{x}}}(t)) - J(v^-_{\tilde{\boldsymbol{x}}}(t)))(v^-_{\tilde{\boldsymbol{x}}}(t) - v^+_{\tilde{\boldsymbol{x}}}(t)) \\ &+ \frac{1}{2}\psi\left(\mathbb{E}\left[v^+_{\tilde{\mathbf{Y}}}\right], \mathbb{E}\left[H_2\left(v^+_{\tilde{\mathbf{Y}}}\right)\right]\right) + \frac{1}{2}\psi\left(\mathbb{E}\left[v^-_{\tilde{\mathbf{Y}}}\right], \mathbb{E}\left[H_2\left(v^-_{\tilde{\mathbf{Y}}}\right)\right]\right).\end{aligned}$$

We are done by the property of Ψ.

□

Differential Equation Approach

Lemma (Maximal Element)

Ψ is a non-empty closed convex set, closed under pointwise maximum, i.e., if $\psi_i \in \Psi$ for $i \in \{1, 2\}$, then $\psi(a, b) = \max(\psi_1(a, b), \psi_2(a, b)) \in \Psi$.

Consequently, the class Ψ has a maximal element $\psi^*(a, b)$ that pointwise dominates all other members of Ψ.

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Coming up: We identify a potential candidate in Ψ, and the finite dimensional mentioned inequalities are those needed to verify its membership in Ψ.

Differential Equation Approach

Using cardinality bound to verify whether $\psi \in \Psi$ yields an inequality of 14 variables.

Definition

Given a four tuple (m_u, m_w, e_u, e_w) satisfying $H_2(m_u) \geq e_u$ and $H_2(m_w) \geq e_w$, let

$$\zeta(m_u, m_w, e_u, e_w) := \inf_{(U,W)\in\mathcal{S}} \frac{1}{2}\mathbb{E}\left[(U-W)(J(W)-J(U))\right],$$

where the set $\mathcal{S}$ is the set of all pairs of random variables $(U, W) \in (0,1)^2$ such that

$$\mathbb{E}[U] = m_u, \mathbb{E}[W] = m_w, \mathbb{E}[H_2(U)] = e_u, \mathbb{E}[H_2(W)] = e_w.$$

It suffices for us to verify

$$\zeta(m_u, m_w, e_u, e_w) \geq \psi\left(\frac{m_u+m_w}{2}, \frac{e_u+e_w}{2}\right) - \frac{1}{2}\psi(m_u, e_u) - \frac{1}{2}\psi(m_w, e_w).$$

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In particular,

$$\zeta(m, 1-m, e, e) \geq \psi\left(\frac{1}{2}, e\right) - \psi(m, e).$$

Differential Equation Approach

Theorem (Origin of our guess)

We have

$$\zeta(m, 1-m, e, e) = \phi\left(\frac{1}{2}, e\right) - \phi(m, e)$$

where $\phi(m, e)$ *is defined as follows: let*

$$\phi(x, y) = \begin{cases} \eta(y) - \frac{y}{r}\eta(r) & H_2(x) > y \\ 0 & H_2(x) \leq y \end{cases}$$

where

$$\eta(x) = (1 - 2H_2^{-1}(x)) \cdot J(H_2^{-1}(x)), \qquad \forall x \in (0, 1].$$

Here $J(x) = \frac{d}{dx}H_2(x)$*, and* $r \in (0, 1]$ *is defined as follows: if* $x = 1/2$*, we set* $r = 1$*; else if* $x \neq \frac{1}{2}$*,* $r \in (0, 1)$ *is the unique solution of the following equation:*

$$\frac{r}{1 - 2H_2^{-1}(r)} = \frac{y}{|1 - 2x|}.$$

Differential Equation Approach

The main conjecture.

Conjecture (Main)

The function ϕ belongs to the class Ψ.

The following lemma addresses the relation between the main conjecture and the mutual information conjecture.

Lemma

If Conjecture (Main) holds, i.e. if $\phi \in \Psi$, then Conjecture (Mutual Information) holds whenever F is balanced, i.e., $Pr(F = -1) = Pr(F = 1) = \frac{1}{2}$.

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Proof.

$$\begin{aligned} g(x) &= \int_{\gamma(0)}^{x} \frac{du}{\phi(\frac{1}{2}, u)} \\ &= \int_{\gamma(0)}^{x} \frac{du}{(1 - 2H_2^{-1}(u)) \cdot J(H_2^{-1}(u))} \\ &= \frac{1}{2} \log \frac{1 - 2H_2^{-1}(\gamma(0))}{1 - 2H_2^{-1}(x)} \end{aligned}$$

Since $g(\gamma(t)) \geq t$, we get $\gamma(t) \geq H_2\left[\frac{1}{2}\left\{1 - e^{-2t}\left[1 - 2H_2^{-1}(\gamma(0))\right]\right\}\right]$.
This implies $1 - I(F; \mathbf{Y}_t) \geq H_2(p_t * H_2^{-1}(1 - I(F; \mathbf{X})))$. For balanced Boolean functions $1 - I(F; \mathbf{X}) = 0$ establishing the conjecture. □

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Remark

It is possible that a similar statement can be made about unbalanced functions as well. However, in this case, the integral of the reciprocal of ϕ does not seem to have a closed form.

Remaining: To check if $\phi \in \Psi$

Define the odd function $L : [0, \frac{1}{2}) \to \mathbb{R}_+$ according to

$$L(u) = \frac{2H_2(u)}{1-2u}, \qquad u \neq \frac{1}{2}$$

and $L(u) = \infty$ when $u = \frac{1}{2}$.

Next, define

$$\kappa(u,w) := (u-w)\frac{J(w)-J(u)}{2} - |u-w| J\left(L^{-1}\left(\frac{H_2(u)+H_2(w)}{|u-w|}\right)\right),$$

for $(u,w) \in (0,1)^2$, $u \neq w$. When $u = w \in [0,1]$, we set $\kappa(u,w) = 0$.

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Conjecture (Properties of $\kappa(u,w)$)

- *for* $u, w \in (0, \frac{1}{2})$, $\kappa(u,w) \leq \kappa(1-u,w)$.
- *for* $l, m \in (0,1)$, $\kappa(H_2^{-1}(l), H_2^{-1}(m))$ *is jointly convex in* l, m

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These three two-variable inequalities (two for convexity) seem to hold by numerical simulations.

To check if $\phi \in \Psi$, ctd

Conjecture

For any $m_u, m_w, e_u, e_w \in [0,1]$ *where* $H_2(m_u) \geq e_u$ *and* $H_2(m_w) \geq e_w$ *we have*

$$\phi\left(\frac{1-|H_2^{-1}(e_u)-H_2^{-1}(e_w)|}{2}, \frac{e_u+e_w}{2}\right) + (H_2^{-1}(e_u)-H_2^{-1}(e_w))\frac{J(H_2^{-1}(e_w))-J(H_2^{-1}(e_u))}{2}$$
$$-\phi\left(\frac{1-|m_u-m_w|}{2}, \frac{e_u+e_w}{2}\right) \geq \phi\left(\frac{m_u+m_w}{2}, \frac{e_u+e_w}{2}\right) - \frac{1}{2}\phi(m_u,e_u) - \frac{1}{2}\phi(m_w,e_w).$$

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Theorem (Take away)

If the properties of κ *hold and the above conjecture holds, then* $\phi \in \Psi$*, and consequently, the mutual information conjecture holds for Balanced functions.*

Remark on the Hellinger

We began working with the Hellinger case.

The whole program was developed for this conjecture

We could establish the value of the equivalent functional $\hat{\zeta}(m, -m, h, h) = \frac{2m^2}{h}$.

This suggested a natural choice for $\hat{\phi}$. However, this choice had counterexamples.

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Therefore, we wanted to know if the program worked for a weaker conjecture (mutual-information) and it seems to.

Conclusion

- We developed a differential-equation-based approach to study the Boolean function conjectures.
- Reduced the checking of the mutual information conjecture to small-dimensional inequalities.

- The origin of the method is from the study of a relay channel problem!!!

Thank You